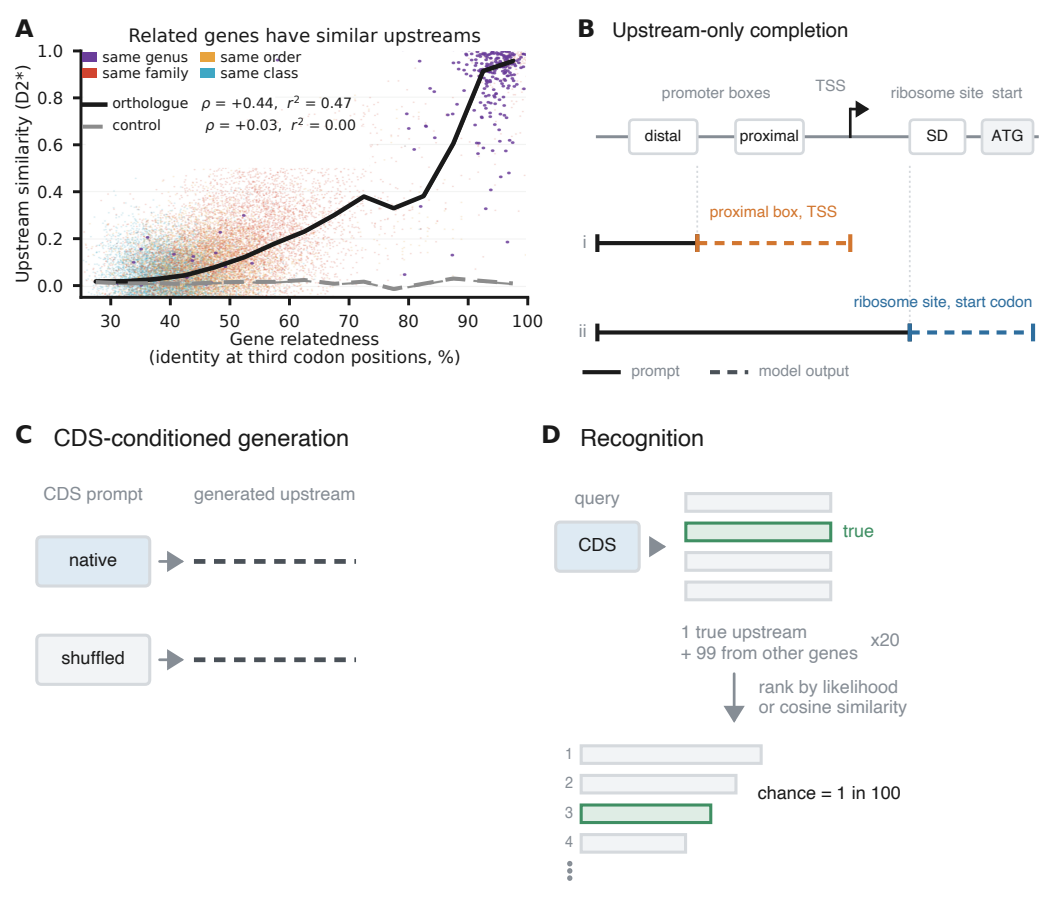

A
Related genes have similar upstreams
same genus
same family
same order
same class
orthologue
control
ρ = +0.44, r² = 0.47
ρ = +0.03, r² = 0.00
Upstream similarity (D2*)
1.0
0.8
0.6
0.4
0.2
0.0
30
40
50
60
70
80
90
100
Gene relatedness
(identity at third codon positions, %)
B
Upstream-only completion
promoter boxes
TSS
ribosome site start
distal
proximal
SD
ATG
proximal box, TSS
i
ribosome site, start codon
ii
prompt
model output
C
CDS-conditioned generation
CDS prompt
generated upstream
native
shuffled
D
Recognition
query
CDS
true
1 true upstream
+ 99 from other genes
x20
rank by likelihood
or cosine similarity
1
2
3
4
chance = 1 in 100